

Not All Scores Are Probabilities: Typed, Calibrated Confidence Contracts for Composed Agent Systems

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Abstract—Composed agent systems pass around many numbers in $[0, 1]$: a model’s self-reported confidence, an embedding similarity, a weighted risk indicator, a posterior mean, a selection propensity. They look interchangeable, and they are routinely thresholded, averaged and multiplied as if they were probabilities. We propose a typed contract. Every emitted score declares its kind (*probability, posterior mean, score, similarity or risk indicator*), its source, its range, whether and against what it is calibrated, and the code that produces it. Five rules then govern combination. We prove the rules’ premises: averaging or taking the maximum across producers is not invariant under monotone reparametrization, so the combined ranking can flip while every input’s ranking is unchanged; products of calibrated, conditionally independent probabilities are calibrated for the conjunction; isotonic regression makes a score calibrated in-sample; and substituting defaults for absent values biases every mean toward the default. Across four systems we find 18 declared scores. Exactly one is a calibrated probability, and none of the 153 cross-producer pairs may be combined numerically. The contract is enforced by a conformance test in every repository, and an isotonic fit reduces an overconfident score’s held-out ECE from 0.277 to 0.018.

I. INTRODUCTION

In the weave systems, ContextWeave rewards a retrieval router with a model’s self-confidence. DeviceWeave executes a command when $\alpha \cdot \text{cosine} + (1 - \alpha) \cdot \text{behaviour} \geq 0.4$. mcp-observatory gates a tool call on a weighted mean of lexical risk indicators. CipherWeave overrides to post-quantum encryption when a max of standardized deviations exceeds 3. Each number lives in $[0, 1]$, or looks like it does, and each means something different. When systems compose, these numbers meet. Nothing in their type says whether 0.7×0.8 or $\frac{1}{2}(0.7 + 0.8)$ means anything.

II. THE CONTRACT

Definition 1 (Score envelope). *A score is emitted as $\langle \text{value}, \text{observed}, \text{name}, \text{kind}, \text{source}, \text{calibrated}, \text{event?}, \text{evidence?}, \text{neutral?} \rangle$. kind is one of:*

- *probability: $P(\text{event})$, and must be calibrated against that event;*
- *posterior_mean: of a stated quantity, carrying its evidence count;*
- *score: ordering-only;*
- *similarity: its scale belongs to the model that produced it, and it declares its neutral point;*
- *risk_indicator: a standardized or weighted indicator, possibly unbounded, carrying its evidence count.*

An unobserved score has value null and observed false.

Rules.

- R1 Only probabilities compose by probability rules.
- R2 A probability must be calibrated, and name against what.
- R3 Never average, add or max scores of different kinds or producers; combine *decisions* instead.
- R4 Carry the evidence count when reliability depends on it.
- R5 Absence is not zero.

Each repository declares its scores in a registry that names the producing function. A conformance test checks the registry against the contract, imports every producer (so a declaration cannot outlive its code), and pins the contract file’s hash, so vendored copies cannot drift. The checker `can_combine(a, b, op)` names the rule that forbids an operation.

III. WHY THE RULES HOLD

Proposition 1 (Cross-producer averages are not ordinal). *Let u, v be scores from different producers, each meaningful only up to a strictly increasing transformation. For any $\alpha \in (0, 1)$ there exist candidates A, B and a strictly increasing f with $\alpha u_A + (1 - \alpha)v_A < \alpha u_B + (1 - \alpha)v_B$ but $\alpha f(u_A) + (1 - \alpha)v_A > \alpha f(u_B) + (1 - \alpha)v_B$.*

Proof. Take $u_A > u_B$ and $v_A < v_B$ with $\alpha(u_A - u_B) < (1 - \alpha)(v_B - v_A)$, so that B ranks higher. Choose f strictly increasing with $f(u_A) - f(u_B) > \frac{1 - \alpha}{\alpha}(v_B - v_A)$, for example a steep enough power. Then A ranks higher. Neither input’s own ordering has changed. $\square$

Concretely, with $\alpha = \frac{1}{2}$, $A = (0.90, 0.10)$ and $B = (0.60, 0.50)$: the raw averages rank B first (0.550 vs. 0.500). Cubing the similarity, a monotone change that could come from simply switching embedding models, ranks A first (0.414 vs. 0.358). This is DeviceWeave’s `final_score` pattern. The same construction applies to max, by choosing f to lift u_A above v_B .

Proposition 2 (Calibrated products are calibrated). *If q_1, q_2 are calibrated probabilities for events E_1, E_2 , i.e. $P(E_i | q_i) = q_i$, and E_1, E_2 are conditionally independent given (q_1, q_2) , with $P(E_i | q_1, q_2) = P(E_i | q_i)$, then $P(E_1 \wedge E_2 | q_1, q_2) = q_1 q_2$.*

Proof. By conditional independence, $P(E_1 \wedge E_2 | q_1, q_2) = P(E_1 | q_1, q_2) P(E_2 | q_1, q_2) = q_1 q_2$. $\square$

TABLE I
DECLARED SCORES BY KIND AND SYSTEM.

System	prob.	post.	score	sim.	risk
ContextWeave	1	1	5	0	0
DeviceWeave	0	0	3	1	0
CipherWeave	0	0	0	0	2
mcp-observatory	0	0	2	0	3
Total (18)	1	1	10	1	5

Both hypotheses are needed, which is why R1 requires the kind to be *probability* and R2 requires calibration. Independence remains the caller’s claim.

Proposition 3 (Isotonic calibration). *Let $\hat{g}$ be the pool-adjacent-violators (PAV) fit to pairs (s_j, o_j) sorted by s . Then $\hat{g}$ is the least-squares non-decreasing fit [1]. On each constant block $\mathcal{B}$ of $\hat{g}$, the fitted value equals the block’s mean outcome, so the calibration error of $\hat{g}$ over its own blocks is zero in-sample.*

Proof. PAV merges adjacent blocks while their means violate monotonicity, and replaces each merged block by its mean. It is standard that this produces the L^2 projection onto the cone of monotone sequences [1]. Each block’s fitted value is by construction the mean of its outcomes, which is the in-sample calibration statement. $\square$

Held out, on 2,500 samples of a synthetic overconfident score (0.9 when right, 0.8 when wrong, 60% right; fitted on a separate 2,500), ECE falls from 0.277 to 0.018 and Brier from 0.258 to 0.108.

Proposition 4 (Defaults bias means). *If a score with mean μ is missing with probability m (independently of its value), and a default κ is substituted, the observed mean is $(1-m)\mu + m\kappa$. Two sources with equal μ but different m then appear different. Dropping missing values recovers μ .*

Proof. This follows from total expectation over the missingness indicator, and from the independence of missingness and value. $\square$

IV. CENSUS

We declared every score emitted by four systems (Table I).

Sources: 8 measurement, 6 heuristic, 2 self-reported, 1 model judge and 1 human. The only calibrated probability is the router’s selection propensity. It is exact by construction, as a Monte Carlo estimate of the chance that Thompson sampling picks the arm. Of the 153 pairs of distinct scores, `can_combine` allows *none* to be averaged, multiplied or maxed. Wherever these systems compose, they must combine decisions (“both gates pass”), not numbers.

V. RELATED WORK

Calibration of classifiers and neural networks is surveyed by Guo et al. [2]. Isotonic and histogram calibration follow Zadrozny and Elkan [3]. Measurement theory’s distinction

between ordinal and interval scales [4] is the formal root of R3.

VI. CONCLUSION

A number in $[0, 1]$ is not a probability until someone has checked it against outcomes. Typing each score, and refusing combinations its type does not license, turns a silent category error into a failing test.

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